Appendix 6 inch-mm

inch		0	1	2	3	4	5	6
Fraction	Decimal	mm						
0	**0.000000**	**0.000**	**25.400**	**50.800**	**76.200**	**101.600**	**127.000**	**152.400**
1/64	0.015625	0.397	25.797	51.197	76.597	101.997	127.397	152.797
1/32	0.031250	0.794	26.194	51.594	76.994	102.394	127.794	153.194
3/64	0.046875	1.191	26.591	51.991	77.391	102.791	128.191	153.591
1/16	**0.062500**	**1.588**	**26.988**	**52.388**	**77.788**	**103.188**	**128.588**	**153.988**
5/64	0.078125	1.984	27.384	52.784	78.184	103.584	128.984	154.384
3/32	0.093750	2.381	27.781	53.181	78.581	103.981	129.381	154.781
7/64	0.109375	2.778	28.178	53.578	78.978	104.378	129.778	155.178
1/8	**0.125000**	**3.175**	**28.575**	**53.975**	**79.375**	**104.775**	**130.175**	**155.575**
9/64	0.140625	3.572	28.972	54.372	79.772	105.172	130.572	155.972
5/32	0.156250	3.969	29.369	54.769	80.169	105.569	130.969	156.369
11/64	0.171875	4.366	29.766	55.166	80.566	105.966	131.366	156.766
3/16	**0.187500**	**4.762**	**30.162**	**55.562**	**80.962**	**106.362**	**131.762**	**157.162**
13/64	0.203125	5.159	30.559	55.959	81.359	106.759	132.159	157.559
7/32	0.218750	5.556	30.956	56.356	81.756	107.156	132.556	157.956
15/64	0.234375	5.953	31.353	56.753	82.153	107.553	132.953	158.353
1/4	**0.250000**	**6.350**	**31.750**	**57.150**	**82.550**	**107.950**	**133.350**	**158.750**
17/64	0.265625	6.747	32.147	57.547	82.947	108.347	133.747	159.147
9/32	0.281250	7.144	32.544	57.944	83.344	108.744	134.144	159.544
19/64	0.296875	7.541	32.941	58.341	83.741	109.141	134.541	159.941
5/16	**0.312500**	**7.938**	**33.338**	**58.738**	**84.138**	**109.538**	**134.938**	**160.338**
21/64	0.328125	8.334	33.734	59.134	84.534	109.934	135.334	160.734
11/32	0.343750	8.731	34.131	59.531	84.931	110.331	135.731	161.131
23/64	0.359375	9.128	34.528	59.928	85.328	110.728	136.128	161.528
3/8	**0.375000**	**9.525**	**34.925**	**60.325**	**85.725**	**111.125**	**136.525**	**161.925**
25/64	0.390625	9.992	35.322	60.722	86.122	111.522	136.922	162.322
13/32	0.406250	10.319	35.719	61.119	86.519	111.919	137.319	162.719
27/64	0.421875	10.716	36.116	61.516	86.916	112.316	137.716	163.116
7/16	**0.437500**	**11.112**	**36.512**	**61.912**	**87.312**	**112.712**	**138.112**	**163.512**
29/64	0.453125	11.509	36.909	62.309	87.709	113.109	138.509	163.909
15/32	0.468750	11.906	37.306	62.706	88.106	113.506	138.906	164.306
31/64	0.484375	12.303	37.703	63.103	88.503	113.903	139.303	164.703
1/2	**0.500000**	**12.700**	**38.100**	**63.500**	**88.900**	**114.300**	**139.700**	**165.100**
33/64	0.515625	13.097	38.497	63.897	89.297	114.697	140.097	165.497
17/32	0.531250	13.494	38.894	64.294	89.694	115.094	140.494	165.894
35/64	0.546875	13.891	39.291	64.691	90.091	115.491	140.891	166.291
9/16	**0.562500**	**14.288**	**39.688**	**65.088**	**90.488**	**115.888**	**141.288**	**166.688**
37/54	0.578125	14.684	40.084	65.484	90.884	116.284	141.684	167.084
19/32	0.593750	15.081	40.481	65.881	91.281	116.681	142.081	167.481
39/64	0.609375	15.478	40.878	66.278	91.678	117.078	142.478	167.878
5/8	**0.625000**	**15.875**	**41.275**	**66.675**	**92.075**	**117.475**	**142.875**	**168.275**
41/64	0.640625	16.272	41.672	67.072	92.472	117.872	143.272	168.672
21/32	0.656250	16.669	42.069	67.469	92.869	118.269	143.669	169.069
43/64	0.671875	17.066	42.466	67.866	93.266	118.666	144.066	169.466
11/16	**0.687500**	**17.462**	**42.862**	**68.262**	**93.662**	**119.062**	**144.462**	**169.862**
45/64	0.703125	17.859	43.259	68.659	94.059	119.459	144.859	170.259
23/32	0.718750	18.256	43.656	69.056	94.456	119.856	145.256	170.656
47/64	0.734375	18.653	44.053	69.453	94.853	120.253	145.653	171.053
3/4	**0.750000**	**19.050**	**44.450**	**69.850**	**95.250**	**120.650**	**146.050**	**171.450**
49/64	0.765625	19.447	44.847	70.247	95.647	121.047	146.447	171.847
25/32	0.781250	19.844	45.244	70.644	96.044	121.444	146.844	172.244
51/64	0.796875	20.241	45.641	71.041	96.441	121.841	147.241	172.641
13/16	**0.812500**	**20.638**	**46.038**	**71.438**	**96.838**	**122.238**	**147.638**	**173.038**
53/64	0.828125	21.034	46.434	71.834	97.234	122.634	148.034	173.434
27/32	0.843750	21.431	46.831	72.231	97.631	123.031	148.431	173.831
55/64	0.859375	21.828	47.228	72.628	98.028	123.428	148.828	174.228
7/8	**0.875000**	**22.225**	**47.625**	**73.025**	**98.425**	**123.825**	**149.225**	**174.625**
57/64	0.890625	22.622	48.022	73.422	98.822	124.222	149.622	175.022
29/32	0.906250	23.019	48.419	73.819	99.219	124.619	150.019	175.419
59/64	0.921875	23.416	48.816	74.216	99.616	125.016	150.416	175.816
15/16	**0.937500**	**23.812**	**49.212**	**74.612**	**100.012**	**125.412**	**150.812**	**176.212**
61/64	0.953125	24.209	49.609	75.009	100.409	125.809	151.209	176.609
31/32	0.968750	24.606	50.006	75.406	100.806	126.206	151.606	177.006
63/64	0.984375	25.003	50.403	75.803	101.203	126.603	152.003	177.403

Conversion Table

1″ =25.4mm

inch		7	8	9	10	11	12
Fraction	Decimal	mm					
0	**0.000000**	**177.800**	**203.200**	**228.600**	**254.000**	**279.400**	**304.800**
1/64	0.015625	178.197	203.597	228.997	254.397	279.797	305.197
1/32	0.031250	178.594	203.994	229.394	254.794	280.194	305.594
3/64	0.046875	178.991	204.391	229.791	255.191	280.591	305.991
1/16	0.062500	**179.388**	**204.788**	**230.188**	**255.588**	**280.988**	**306.388**
5/64	0.078125	179.784	205.184	230.584	255.984	281.384	306.784
3/32	0.093750	180.181	205.581	230.981	256.381	281.781	307.181
7/64	0.109375	180.578	205.978	231.378	256.778	282.178	307.578
1/8	**0.125000**	**180.975**	**206.375**	**231.775**	**257.175**	**282.575**	**307.975**
9/64	0.140625	181.372	206.772	232.172	257.572	282.972	308.372
5/32	0.156250	181.769	207.169	232.569	257.969	283.369	308.769
11/64	0.171875	181.166	207.566	232.966	258.366	283.766	309.166
3/16	**0.187500**	**182.562**	**207.962**	**233.362**	**258.762**	**284.162**	**309.562**
13/64	0.203125	182.959	208.359	233.759	259.159	284.559	309.959
7/32	0.218750	183.356	208.756	234.156	259.556	284.956	310.356
15/64	0.234375	183.753	209.153	234.553	259.953	285.353	310.753
1/4	**0.250000**	**184.150**	**209.550**	**234.950**	**260.350**	**285.750**	**311.150**
17/64	0.265625	184.547	209.947	235.347	260.747	286.147	311.547
9/32	0.281250	184.944	210.344	235.744	261.144	286.544	311.944
19/64	0.296875	185.341	210.741	236.141	261.541	286.941	312.341
5/16	**0.312500**	**185.738**	**211.138**	**236.538**	**261.938**	**287.338**	**312.738**
21/64	0.328125	186.134	211.534	236.934	262.334	287.734	313.134
11/32	0.343750	186.531	211.931	237.331	262.731	288.131	313.531
23/64	0.359375	186.928	212.328	237.728	263.128	288.528	313.928
3/8	**0.375000**	**187.325**	**212.725**	**238.125**	**263.525**	**288.925**	**314.325**
25/64	0.390625	187.722	213.122	238.522	263.922	289.322	314.722
13/32	0.406250	188.119	213.519	238.919	264.319	289.719	315.119
27/64	0.421875	188.516	213.916	239.316	264.716	290.116	315.516
7/16	**0.437500**	**188.912**	**214.312**	**239.712**	**265.112**	**290.512**	**315.912**
29/64	0.453125	189.309	214.709	240.109	265.509	290.909	316.309
15/32	0.468750	189.706	215.106	240.506	265.906	291.306	316.706
31/64	0.484375	190.103	215.503	240.903	266.303	291.703	317.103
1/2	**0.500000**	**190.500**	**215.900**	**241.300**	**266.700**	**292.100**	**317.500**
33/64	0.515625	190.897	216.297	241.697	267.097	292.497	317.897
17/32	0.531250	191.294	216.694	242.094	267.494	292.894	318.294
35/64	0.546875	191.691	217.091	242.491	267.891	293.291	318.691
9/16	**0.562500**	**192.088**	**217.488**	**242.888**	**268.288**	**293.688**	**319.088**
37/54	0.578125	192.484	217.884	243.284	268.684	294.084	319.484
19/32	0.593750	192.881	218.281	243.681	269.081	294.481	319.881
39/64	0.609375	193.278	218.678	244.078	269.478	294.878	320.278
5/8	**0.625000**	**193.675**	**219.075**	**244.475**	**269.875**	**295.275**	**320.675**
41/64	0.640625	194.072	219.472	244.872	270.272	295.672	321.072
21/32	0.656250	194.469	219.869	245.269	270.669	296.069	321.469
43/64	0.671875	194.866	220.266	245.666	271.066	296.466	321.866
11/16	**0.687500**	**195.262**	**220.662**	**246.062**	**271.462**	**296.862**	**322.262**
45/64	0.703125	195.659	221.059	246.459	271.859	297.259	322.659
23/32	0.718750	196.056	221.456	246.856	272.256	297.656	323.056
47/64	0.734375	196.453	221.853	247.253	272.653	298.053	323.453
3/4	**0.750000**	**196.850**	**222.250**	**247.650**	**273.050**	**298.450**	**323.850**
49/64	0.765625	197.247	222.647	248.047	273.447	298.847	324.247
25/32	0.781250	197.644	223.044	248.444	273.844	299.244	324.644
51/64	0.796875	198.041	223.441	248.841	274.241	299.641	325.041
13/16	**0.812500**	**198.438**	**223.838**	**249.238**	**274.638**	**300.038**	**325.438**
53/64	0.828125	198.834	224.234	249.634	275.034	300.434	325.834
27/32	0.843750	199.231	224.631	250.031	275.431	300.831	326.231
55/64	0.859375	199.628	225.028	250.428	275.828	301.228	326.628
7/8	**0.875000**	**200.025**	**225.425**	**250.825**	**276.225**	**301.625**	**327.050**
57/64	0.890625	200.422	225.822	251.222	276.622	302.022	327.422
29/32	0.906250	200.819	226.219	251.619	277.019	302.419	327.819
59/64	0.921875	201.216	226.616	252.016	277.416	302.816	328.216
15/16	**0.937500**	**201.612**	**227.012**	**252.412**	**277.812**	**303.212**	**328.612**
61/64	0.953125	202.009	227.409	252.809	278.209	303.609	329.009
31/32	0.968750	202.406	227.806	253.206	278.606	304.006	329.406
63/64	0.984375	202.803	228.203	253.603	279.003	304.403	329.803

Appendix 7 Hardness Conversion Table (Reference)

Rockwell C Scale Hardness (1 471N) {150kgf}	Vickers Hardness	Brinell Hardness		Rockwell Hardness		Shore Hardness
		Standard Ball	Tungsten Carbide Ball	A Scale Load 588.4N {60kgf} Brale Indenter	B Scale Load 980.7N {100kgf} 1.588mm (1/16in) Ball	
68	940	—	—	85.6	—	97
67	900	—	—	85.0	—	95
66	865	—	—	84.5	—	92
65	832	—	739	83.9	—	91
64	800	—	722	83.4	—	88
63	772	—	705	82.8	—	87
62	746	—	688	82.3	—	85
61	720	—	670	81.8	—	83
60	697	—	654	81.2	—	81
59	674	—	634	80.7	—	80
58	653	—	615	80.1	—	78
57	633	—	595	79.6	—	76
56	613	—	577	79.0	—	75
55	595	—	560	78.5	—	74
54	577	—	543	78.0	—	72
53	560	—	525	77.4	—	71
52	544	500	512	76.8	—	69
51	528	487	496	76.3	—	68
50	513	475	481	75.9	—	67
49	498	464	469	75.2	—	66
48	484	451	455	74.7	—	64
47	471	442	443	74.1	—	63
46	458	432	432	73.6	—	62
45	446	421	421	73.1	—	60
44	434	409	409	72.5	—	58
43	423	400	400	72.0	—	57
42	412	390	390	71.5	—	56
41	402	381	381	70.9	—	55
40	392	371	371	70.4	—	54
39	382	362	362	69.9	—	52
38	372	353	353	69.4	—	51
37	363	344	344	68.9	—	50
36	354	336	336	68.4	(109.0)	49
35	345	327	327	67.9	(108.5)	48
34	336	319	319	67.4	(108.0)	47
33	327	311	311	66.8	(107.5)	46
32	318	301	301	66.3	(107.0)	44
31	310	294	294	65.8	(106.0)	43
30	302	286	286	65.3	(105.5)	42
29	294	279	279	64.7	(104.5)	41
28	286	271	271	64.3	(104.0)	41
27	279	264	264	63.8	(103.0)	40
26	272	258	258	63.3	(102.5)	38
25	266	253	253	62.8	(101.5)	38
24	260	247	247	62.4	(101.0)	37
23	254	243	243	62.0	100.0	36
22	248	237	237	61.5	99.0	35
21	243	231	231	61.0	98.5	35
20	238	226	226	60.5	97.8	34
(18)	230	219	219	—	96.7	33
(16)	222	212	212	—	95.5	32
(14)	213	203	203	—	93.9	31
(12)	204	194	194	—	92.3	29
(10)	196	187	187	—	90.7	28
(8)	188	179	179	—	89.5	27
(6)	180	171	171	—	87.1	26
(4)	173	165	165	—	85.5	25
(2)	166	158	158	—	83.5	24
(0)	160	152	152	—	81.7	24

Appendix 8 Physical and Mechanical Properties of Materials

Materials	Specific Gravity	Coefficient of Linear Expansion (0° to 100°C)	Hardness (Brinell)	Modulus of Linear Elasticity (MPa) {kgf/mm²}	Tensile Strength (MPa) {kgf/mm²}	Yield Point (MPa) {kgf/mm²}	Elongation (%)
Bearing Steel (hardened)	7.83	12.5×10^{-6}	650 to 740	208 000 {21 200}	1 570 to 1 960 {160 to 200}	—	—
Martensitic Stainless Steel SUS 440C	7.68	10.1×10^{-6}	580	200 000 {20 400}	1 960 {200}	1 860 {190}	—
Mild Steel (C=0.12 to 0.20%)	7.86	11.6×10^{-6}	100 to 130	206 000 {21 000}	373 to 471 {38 to 48}	216 to 294 {22 to 30}	24 to 36
Hard Steel (C=0.3 to 0.5%)	7.84	11.3×10^{-6}	160 to 200	206 000 {21 000}	539 to 686 {55 to 70}	333 to 451 {34 to 46}	14 to 26
Austenitic Stainless Steel SUS 304	8.03	16.3×10^{-6}	150	193 000 {19 700}	588 {60}	245 {25}	60
Cast Iron — Gray Iron FC200	7.3	10.4×10^{-6}	Less than 223	98 100 {10 000}	More than 200 {20}	—	—
Cast Iron — Spheroidal graphite Iron FCD400	7.0	11.7×10^{-6}	Less than 201		More than 400 {41}	—	More than 12
Aluminum	2.69	23.7×10^{-6}	15 to 26	70 600 {7 200}	78 {8}	34 {3.5}	35
Zinc	7.14	31×10^{-6}	30 to 60	92 200 {9 400}	147 {15}	—	30 to 40
Copper	8.93	16.2×10^{-6}	50	123 000 {12 500}	196 {20}	69 {7}	15 to 20
Brass (Annealed)	8.5	19.1×10^{-6}	45	103 000 {10 500}	294 to 343 {30 to 35}	—	65 to 75
Brass (Machined)			85 to 130		363 to 539 {37 to 55}		15 to 50

Remarks The hardness of hardened bearing steel and martensitic stainless steel is usually expressed using the Rockwell C Scale, but for comparison, it is converted into Brinell hardness.

Appendix 9 Tolerances for Shaft Diameters

Units: μm

Diameter Classification (mm)		Single Plane Mean Bore Deviation (Normal) Δ_{dmp}	a	c	d	e	f		g		h						js		j		k		m		n		p		r		Diameter Classification (mm)	
over	incl		a13	c12	d6	e6	f5	f6	g5	g6	h5	h6	h7	h8	h9	h10	js5	js6	j5	j6	k5	k6	m5	m6	n5	n6	p5	p6	r6	r7	over	incl
3	6	0 − 8	−270 −450	− 70 −190	− 30 − 38	− 20 − 28	−10 −15	− 10 − 18	− 4 − 9	− 4 −12	0 − 5	0 − 8	0 −12	0 −18	0 − 30	0 − 48	±2.5	± 4	+ 3 − 2	+ 6 − 2	+ 6 + 1	+ 9 + 1	+ 9 + 4	+12 + 4	+13 + 8	+16 + 8	+17 +12	+ 20 + 12	+ 23 + 15	+ 27 + 15	3	6
6	10	0 − 8	−280 −500	− 80 −230	− 40 − 49	− 25 − 34	−13 −19	− 13 − 22	− 5 −11	− 5 −14	0 − 6	0 − 9	0 −15	0 −22	0 − 36	0 − 58	± 3	±4.5	+ 4 − 2	+ 7 − 2	+ 7 + 1	+10 + 1	+12 + 6	+15 + 6	+16 +10	+19 +10	+21 +15	+ 24 + 15	+ 28 + 19	+ 34 + 19	6	10
10	14	0 − 8	−290 −560	− 95 −275	− 50 − 61	− 32 − 43	−16 −24	− 16 − 27	− 6 −14	− 6 −17	0 − 8	0 −11	0 −18	0 −27	0 − 43	0 − 70	± 4	±5.5	+ 5 − 3	+ 8 − 3	+ 9 + 1	+12 + 1	+15 + 7	+18 + 7	+20 +12	+23 +12	+26 +18	+ 29 + 18	+ 34 + 23	+ 41 + 23	10	14
14	18																														14	18
18	24	0 −10	−300 −630	−110 −320	− 65 − 78	− 40 − 53	−20 −29	− 20 − 33	− 7 −16	− 7 −20	0 − 9	0 −13	0 −21	0 −33	0 − 52	0 − 84	±4.5	±6.5	+ 5 − 4	+ 9 − 4	+11 + 2	+15 + 2	+17 + 8	+21 + 8	+24 +15	+28 +15	+31 +22	+ 35 + 22	+ 41 + 28	+ 49 + 28	18	24
24	30																														24	30
30	40	0 −12	− 310 − 700	− 120 − 370	− 80 − 96	− 50 − 66	−25 −36	− 25 − 41	− 9 −20	− 9 −25	0 −11	0 −16	0 −25	0 −39	0 − 62	0 −100	±5.5	± 8	+ 6 − 5	+11 − 5	+13 + 2	+18 + 2	+20 + 9	+25 + 9	+28 +17	+33 +17	+37 +26	+ 42 + 26	+ 50 + 34	+ 59 + 34	30	40
40	50		− 320 − 710	− 130 − 380																											40	50
50	65	0 −15	− 340 − 800	− 140 − 440	−100 −119	− 60 − 79	−30 −43	− 30 − 49	−10 −23	−10 −29	0 −13	0 −19	0 −30	0 −46	0 − 74	0 −120	±6.5	±9.5	+ 6 − 7	+12 − 7	+15 + 2	+21 + 2	+24 +11	+30 +11	+33 +20	+39 +20	+45 +32	+ 51 + 32	+ 60 + 41	+ 71 + 41	50	65
65	80		− 360 − 820	− 150 − 450																									+ 62 + 43	+ 73 + 43	65	80
80	100	0 −20	− 380 − 920	− 170 − 520	−120 −142	− 72 − 94	−36 −51	− 36 − 58	−12 −27	−12 −34	0 −15	0 −22	0 −35	0 −54	0 − 87	0 −140	±7.5	±11	+ 6 − 9	+13 − 9	+18 + 3	+25 + 3	+28 +13	+35 +13	+38 +23	+45 +23	+52 +37	+ 59 + 37	+ 73 + 51	+ 86 + 51	80	100
100	120		− 410 − 959	− 180 − 530																									+ 76 + 54	+ 89 + 54	100	120
120	140	0 −25	− 460 −1 090	− 200 − 600	−145 −170	− 85 −110	−43 −61	− 43 − 68	−14 −32	−14 −39	0 −18	0 −25	0 −40	0 −63	0 −100	0 −160	± 9	±12.5	+ 7 −11	+14 −11	+21 + 3	+28 + 3	+33 +15	+40 +15	+45 +27	+52 +27	+61 +43	+ 68 + 43	+ 88 + 63	+103 + 63	120	140
140	160		− 520 −1 150	− 210 − 610																									+ 90 + 65	+105 + 65	140	160
160	180		− 580 −1 210	− 230 − 630																									+ 93 + 68	+108 + 68	160	180
180	200	0 −30	− 660 −1 380	− 240 − 700	−170 −199	−100 −129	−50 −70	− 50 − 79	−15 −35	−15 −44	0 −20	0 −29	0 −46	0 −72	0 −115	0 −185	±10	±14.5	+ 7 −13	+16 −13	+24 + 4	+33 + 4	+37 +17	+46 +17	+51 +31	+60 +31	+70 +50	+ 79 + 50	+106 + 77	+123 + 77	180	200
200	225		− 740 −1 460	− 260 − 720																									+109 + 80	+126 + 80	200	225
225	250		− 820 −1 540	− 280 − 740																									+113 + 84	+130 + 84	225	250
250	280	0 −35	− 920 −1 730	− 300 − 820	−190 −222	−110 −142	−56 −79	− 56 − 88	−17 −40	−17 −49	0 −23	0 −32	0 −52	0 −81	0 −130	0 −210	±11.5	±16	+ 7 −16	±16	+27 + 4	+36 + 4	+43 +20	+52 +20	+57 +34	+66 +34	+79 +56	+ 88 + 56	+126 + 94	+146 + 94	250	280
280	315		−1 050 −1 860	− 330 − 850																									+130 + 98	+150 + 98	280	315
315	355	0 −40	−1 200 −2 090	− 360 − 930	−210 −246	−125 −161	−62 −87	− 62 − 98	−18 −43	−18 −54	0 −25	0 −36	0 −57	0 −89	0 −140	0 −230	±12.5	±18	+ 7 −18	±18	+29 + 4	+40 + 4	+46 +21	+57 +21	+62 +37	+73 +37	+87 +62	+ 98 + 62	+144 +108	+165 +108	315	355
355	400		−1 350 −2 240	− 400 − 970																									+150 +114	+171 +114	355	400
400	450	0 −45	−1 500 −2 470	− 440 −1 070	−230 −270	−135 −175	−68 −95	− 68 −108	−20 −47	−20 −60	0 −27	0 −40	0 −63	0 −97	0 −155	0 −250	±13.5	±20	+ 7 −20	±20	+32 + 5	+45 + 5	+50 +23	+63 +23	+67 +40	+80 +40	+95 +68	+108 + 68	+166 +126	+189 +126	400	450
450	500		−1 650 −2 620	− 480 −1 110																									+172 +132	+195 +132	450	500

Appendix 10 Tolerances for Housing Bore Diameters

Units: μm

Diameter Classification (mm)		Single Plane Mean O. D. Deviation (Normal) Δ_{Dmp}	E		F			G		H					
over	incl		E10	E11	F6	F7	F8	G6	G7	H5	H6	H7	H8	H9	H10
3	6	0 − 8	+ 68 + 20	+ 95 + 20	+ 18 + 10	+ 22 + 10	+ 28 + 10	+12 + 4	+16 + 4	+ 5 0	+ 8 0	+12 0	+18 0	+ 30 0	+ 48 0
6	10	0 − 8	+ 83 + 25	+115 + 25	+ 22 + 13	+ 28 + 13	+ 35 + 13	+14 + 5	+20 + 5	+ 6 0	+ 9 0	+15 0	+22 0	+ 36 0	+ 58 0
10	14	0 − 8	+102 + 32	+142 + 32	+ 27 + 16	+ 34 + 16	+ 43 + 16	+17 + 6	+24 + 6	+ 8 0	+11 0	+18 0	+27 0	+ 43 0	+ 70 0
14	18														
18	24	0 − 9	+124 + 40	+170 + 40	+ 33 + 20	+ 41 + 20	+ 53 + 20	+20 + 7	+28 + 7	+ 9 0	+13 0	+21 0	+33 0	+ 52 0	+ 84 0
24	30														
30	40	0 −11	+150 + 50	+210 + 50	+ 41 + 25	+ 50 + 25	+ 64 + 25	+25 + 9	+34 + 9	+11 0	+16 0	+25 0	+39 0	+ 62 0	+100 0
40	50														
50	65	0 −13	+180 + 60	+250 + 60	+ 49 + 30	+ 60 + 30	+ 76 + 30	+29 +10	+40 +10	+13 0	+19 0	+30 0	+46 0	+ 74 0	+120 0
65	80														
80	100	0 −15	+212 + 72	+292 + 72	+ 58 + 36	+ 71 + 36	+ 90 + 36	+34 +12	+47 +12	+15 0	+22 0	+35 0	+54 0	+ 87 0	+140 0
100	120														
120	140	0 −18 ($D \leqq 150$)	+245 + 85	+335 + 85	+ 68 + 43	+ 83 + 43	+106 + 43	+39 +14	+54 +14	+18 0	+25 0	+40 0	+63 0	+100 0	+160 0
140	160														
160	180	0 −25 ($D > 150$)													
180	200	0 −30	+285 +100	+390 +100	+ 79 + 50	+ 96 + 50	+122 + 50	+44 +15	+61 +15	+20 0	+29 0	+46 0	+72 0	+115 0	+185 0
200	225														
225	250														
250	280	0 −35	+320 +110	+430 +110	+ 88 + 56	+108 + 56	+137 + 56	+46 +17	+69 +17	+23 0	+32 0	+52 0	+81 0	+130 0	+210 0
280	315														
315	355	0 −40	+355 +125	+485 +125	+ 98 + 62	+119 + 62	+151 + 62	+54 +18	+75 +18	+25 0	+36 0	+57 0	+89 0	+140 0	+230 0
355	400														
400	450	0 −45	+385 +135	+535 +135	+108 + 68	+131 + 68	+165 + 68	+60 +20	+83 +20	+27 0	+40 0	+63 0	+97 0	+155 0	+250 0
450	500														

JS		J		K		M		N		P		R		Diameter Classification (mm)	
JS6	JS7	J6	J7	K6	K7	M6	M7	N6	N7	P6	P7	R6	R7	over	incl
± 4	± 6	+ 5 − 3	± 6	+ 2 − 6	+ 3 − 9	− 1 − 9	0 −12	− 5 −13	− 4 −16	− 9 −17	− 8 − 20	− 12 − 20	− 11 − 23	3	6
±4.5	± 7	+ 5 − 4	+ 8 − 7	+ 2 − 7	+ 5 −10	− 3 −12	0 −15	− 7 −16	− 4 −19	−12 −21	− 9 − 24	− 16 − 25	− 13 − 28	6	10
±5.5	± 9	+ 6 − 5	+10 − 8	+ 2 − 9	+ 6 −12	− 4 −15	0 −18	− 9 −20	− 5 −23	−15 −26	− 11 − 29	− 20 − 31	− 16 − 34	10	14
														14	18
±6.5	±10.5	+ 8 − 5	+12 − 9	+ 2 −11	+ 6 −15	− 4 −17	0 −21	−11 −24	− 7 −28	−18 −31	− 14 − 35	− 24 − 37	− 20 − 41	18	24
														24	30
± 8	±12.5	+10 − 6	+14 −11	+ 3 −13	+ 7 −18	− 4 −20	0 −25	−12 −28	− 8 −33	−21 −37	− 17 − 42	− 29 − 45	− 25 − 50	30	40
														40	50
±9.5	±15	+13 − 6	+18 −12	+ 4 −15	+ 9 −21	− 5 −24	0 −30	−14 −33	− 9 −39	−26 −45	− 21 − 51	− 35 − 54	− 30 − 60	50	65
												− 37 − 56	− 32 − 62	65	80
±11	±17.5	+16 − 6	+22 −13	+ 4 −18	+10 −25	− 6 −28	0 −35	−16 −38	−10 −45	−30 −52	− 24 − 59	− 44 − 66	− 38 − 73	80	100
												− 47 − 69	− 41 − 76	100	120
±12.5	±20	+18 − 7	+26 −14	+ 4 −21	+12 −28	− 8 −33	0 −40	−20 −45	−12 −52	−36 −61	− 28 − 68	− 56 − 81	− 48 − 88	120	140
												− 58 − 83	− 50 − 90	140	160
												− 61 − 86	− 53 − 93	160	180
±14.5	±23	+22 − 7	+30 −16	+ 5 −24	+13 −33	− 8 −37	0 −46	−22 −51	−14 −60	−41 −70	− 33 − 79	− 68 − 97	− 60 −106	180	200
												− 71 −100	− 63 −109	200	225
												− 75 −104	− 67 −113	225	250
±16	±26	+25 − 7	+36 −16	+ 5 −27	+16 −36	− 9 −41	0 −52	−25 −57	−14 −66	−47 −79	− 36 − 88	− 85 −117	− 74 −126	250	280
												− 89 −121	− 78 −130	280	315
±18	±28.5	+29 − 7	+39 −18	+ 7 −29	+17 −40	−10 −46	0 −57	−26 −62	−16 −73	−51 −87	− 41 − 98	− 97 −133	− 87 −144	315	355
												−103 −139	− 93 −150	355	400
±20	±31.5	+33 − 7	+43 −20	+ 8 −32	+18 −45	−10 −50	0 −63	−27 −67	−17 −80	−55 −95	− 45 −108	−113 −153	−103 −166	400	450
												−119 −159	−109 −172	450	500

Appendix 11 Values of Standard Tolerance Grades IT

Basic Size (mm) over	Basic Size (mm) incl	Standard Tolerance Grades IT1 (μm)	IT2	IT3	IT4	IT5	IT6	IT7	IT8	IT9	IT10	IT11
—	3	0.8	1.2	2	3	4	6	10	14	25	40	60
3	6	1	1.5	2.5	4	5	8	12	18	30	48	75
6	10	1	1.5	2.5	4	6	9	15	22	36	58	90
10	18	1.2	2	3	5	8	11	18	27	43	70	110
18	30	1.5	2.5	4	6	9	13	21	33	52	84	130
30	50	1.5	2.5	4	7	11	16	25	39	62	100	160
50	80	2	3	5	8	13	19	30	46	74	120	190
80	120	2.5	4	6	10	15	22	35	54	87	140	220
120	180	3.5	5	8	12	18	25	40	63	100	160	250
180	250	4.5	7	10	14	20	29	46	72	115	185	290
250	315	6	8	12	16	23	32	52	81	130	210	320
315	400	7	9	13	18	25	36	57	89	140	230	360
400	500	8	10	15	20	27	40	63	97	155	250	400
500	630	9	11	16	22	32	44	70	110	175	280	440
630	800	10	13	18	25	36	50	80	125	200	320	500
800	1 000	11	15	21	28	40	56	90	140	230	360	560
1 000	1 250	13	18	24	33	47	66	105	165	260	420	660
1 250	1 600	15	21	29	39	55	78	125	195	310	500	780
1 600	2 000	18	25	35	46	65	92	150	230	370	600	920
2 000	2 500	22	30	41	55	78	110	175	280	440	700	1 100
2 500	3 150	26	36	50	68	96	135	210	330	540	860	1 350

Grades IT12	IT13	IT14	IT15	IT16	IT17	IT18	Basic Size (mm) over	incl
Tolerances (mm)								
0.10	0.14	0.25	0.40	0.60	1.00	1.40	—	3
0.12	0.18	0.30	0.48	0.75	1.20	1.80	3	6
0.15	0.22	0.36	0.58	0.90	1.50	2.20	6	10
0.18	0.27	0.43	0.70	1.10	1.80	2.70	10	18
0.21	0.33	0.52	0.84	1.30	2.10	3.30	18	30
0.25	0.39	0.62	1.00	1.60	2.50	3.90	30	50
0.30	0.46	0.74	1.20	1.90	3.00	4.60	50	80
0.35	0.54	0.87	1.40	2.20	3.50	5.40	80	120
0.40	0.63	1.00	1.60	2.50	4.00	6.30	120	180
0.46	0.72	1.15	1.85	2.90	4.60	7.20	180	250
0.52	0.81	1.30	2.10	3.20	5.20	8.10	250	315
0.57	0.89	1.40	2.30	3.60	5.70	8.90	315	400
0.63	0.97	1.55	2.50	4.00	6.30	9.70	400	500
0.70	1.10	1.75	2.80	4.40	7.00	11.00	500	630
0.80	1.25	2.00	3.20	5.00	8.00	12.50	630	800
0.90	1.40	2.30	3.60	5.60	9.00	14.00	800	1 000
1.05	1.65	2.60	4.20	6.60	10.50	16.50	1 000	1 250
1.25	1.95	3.10	5.00	7.80	12.50	19.50	1 250	1 600
1.50	2.30	3.70	6.00	9.20	15.00	23.00	1 600	2 000
1.75	2.80	4.40	7.00	11.00	17.50	28.00	2 000	2 500
2.10	3.30	5.40	8.60	13.50	21.00	33.00	2 500	3 150

Remarks 1. Standard tolerance grades IT14 to IT18 shall not be used for basic sizes less than or equal to 1 mm.
2. Values for standard tolerance grades IT1 to IT5 for basic sizes over 500 mm are included for experimental use.

Appendix 12 Speed Factor f_n

Ball Bearings $f_n = (0.03n)^{-1/3}$
Roller Bearings $f_n = (0.03n)^{-3/10}$

Speed n (min^{-1})	Speed Factor f_n	
	Ball Bearings	Roller Bearings
10	1.49	1.44
11	1.45	1.39
12	1.41	1.36
13	1.37	1.33
14	1.34	1.30
15	1.30	1.27
16	1.28	1.25
17	1.25	1.22
18	1.23	1.20
19	1.21	1.18
20	1.19	1.17
21	1.17	1.15
22	1.15	1.13
23	1.13	1.12
24	1.12	1.10
25	1.10	1.09
26	1.09	1.08
27	1.07	1.07
28	1.06	1.05
29	1.05	1.04
30	1.04	1.03
31	1.02	1.02
32	1.01	1.01
33.3	**1.00**	**1.00**
34	0.993	0.994
36	0.975	0.977
38	0.957	0.961
40	0.941	0.947
42	0.926	0.933
44	0.912	0.920
46	0.898	0.908
48	0.886	0.896
50	0.874	0.885
55	0.846	0.861
60	0.822	0.838
65	0.800	0.818
70	0.781	0.800
75	0.763	0.784
80	0.747	0.769
85	0.732	0.755
90	0.718	0.742
95	0.705	0.730
100	0.693	0.719
110	0.672	0.699
120	0.652	0.681
130	0.635	0.665
140	0.620	0.650
150	0.606	0.637
160	0.593	0.625
170	0.581	0.613
180	0.570	0.603
190	0.560	0.593
200	0.550	0.584
220	0.533	0.568
240	0.518	0.553
260	0.504	0.540
280	0.492	0.528
300	0.481	0.517
320	0.471	0.507
340	0.461	0.498
360	0.452	0.490
380	0.444	0.482
400	0.437	0.475
420	0.430	0.468
440	0.423	0.461
460	0.417	0.455
480	0.411	0.449
500	0.405	0.444
550	0.393	0.431
600	0.382	0.420
650	0.372	0.410
700	0.362	0.401
750	0.354	0.393
800	0.347	0.385
850	0.340	0.378
900	0.333	0.372
950	0.327	0.366
1 000	0.322	0.360
1 050	0.317	0.355
1 100	0.312	0.350
1 150	0.307	0.346
1 200	0.303	0.341
1 250	0.299	0.337
1 300	0.295	0.333
1 400	0.288	0.326
1 500	0.281	0.319
1 600	0.275	0.313
1 700	0.270	0.307
1 800	0.265	0.302
1 900	0.260	0.297
2 000	0.255	0.293
2 100	0.251	0.289
2 200	0.247	0.285
2 300	0.244	0.281
2 400	0.240	0.277
2 500	0.237	0.274
2 600	0.234	0.271
2 700	0.231	0.268
2 800	0.228	0.265
2 900	0.226	0.262
3 000		0.259
3 200		0.254
3 400		0.250
3 600		0.245
3 800		0.242
4 000		0.238
4 200		0.234
4 400		0.231
4 600		0.228
4 800		0.225
5 000		0.222
5 200		0.220
5 400		0.217
5 600		0.215
5 800		0.213
6 000		0.211
6 200		0.209
6 400		0.207
6 600		0.205
6 800		0.203
7 000		0.201
7 200		0.199
7 400		0.198
7 600		0.196
7 800		0.195
8 000		0.193
8 500		0.190
9 000		0.186
9 500		0.183
10 000		0.181
11 000		0.176
12 000		0.171
13 000		0.167
14 000		0.163
15 000		0.160
16 000		0.157
17 000		0.154
18 000		0.151
19 000		0.149
20 000		0.147
22 000		0.143
24 000		0.139
26 000		0.136
28 000		0.133
30 000		0.130
32 000		0.127
34 000		0.125
36 000		0.123
38 000		0.121
40 000		0.119

Appendix 13 Fatigue Life Factor f_h and Fatigue Life $L \cdot L_h$

Ball Bearings $L = (C/P)^3$ $L_h = 500 f_h^3$
Roller Bearings $L = (C/P)^{10/3}$ $L_h = 500 f_h^{10/3}$

C/P or f_h	Ball Bearing Life		Roller Bearing Life	
	L (10^6 rev)	L_h (h)	L (10^6 rev)	L_h (h)
0.70	0.34	172	0.30	152
0.75	0.42	211	0.38	192
0.80	0.51	256	0.48	238
0.85	0.61	307	0.58	291
0.90	0.73	365	0.70	352
0.95	0.86	429	0.84	421
1.00	**1.00**	**500**	**1.00**	**500**
1.05	1.16	579	1.18	588
1.10	1.33	665	1.37	687
1.15	1.52	760	1.59	797
1.20	1.73	864	1.84	918
1.25	1.95	977	2.10	1 050
1.30	2.20	1 100	2.40	1 200
1.35	2.46	1 230	2.72	1 360
1.40	2.74	1 370	3.07	1 530
1.45	3.05	1 520	3.45	1 730
1.50	3.38	1 690	3.86	1 930
1.55	3.72	1 860	4.31	2 150
1.60	4.10	2 050	4.79	2 400
1.65	4.49	2 250	5.31	2 650
1.70	4.91	2 460	5.86	2 930
1.75	5.36	2 680	6.46	3 230
1.80	5.83	2 920	7.09	3 550
1.85	6.33	3 170	7.77	3 890
1.90	6.86	3 430	8.50	4 250
1.95	7.41	3 710	9.26	4 630
2.00	8.00	4 000	10.1	5 040
2.05	8.62	4 310	10.9	5 470
2.10	9.26	4 630	11.9	5 930
2.15	9.94	4 970	12.8	6 410
2.20	10.6	5 320	13.8	6 920
2.25	11.4	5 700	14.9	7 460
2.30	12.2	6 080	16.1	8 030
2.35	13.0	6 490	17.3	8 630
2.40	13.8	6 910	18.5	9 250
2.45	14.7	7 350	19.8	9 910
2.50	15.6	7 810	21.2	10 600
2.55	16.6	8 290	22.7	11 300
2.60	17.6	8 790	24.2	12 100
2.65	18.6	9 300	25.8	12 900
2.70	19.7	9 840	27.4	13 700
2.75	20.8	10 400	29.1	14 600
2.80	22.0	11 000	30.9	15 500
2.85	23.1	11 600	32.8	16 400
2.90	24.4	12 200	34.8	17 400
2.95	25.7	12 800	36.8	18 400
3.00	27.0	13 500	38.9	19 500
3.05	28.4	14 200	41.1	20 600
3.10	29.8	14 900	43.4	21 700
3.15	31.3	15 600	45.8	22 900
3.20	32.8	16 400	48.3	24 100
3.25	34.3	17 200	50.8	25 400
3.30	35.9	18 000	53.5	26 800
3.35	37.6	18 800	56.3	28 100
3.40	39.3	19 700	59.1	29 600
3.45	41.1	20 500	62.0	31 000
3.50	42.9	21 400	65.1	32 500
3.55	44.7	22 400	68.2	34 100
3.60	46.7	23 300	71.5	35 800
3.65	48.6	24 300	74.9	37 400
3.70	50.7	25 300	78.3	39 200
3.75	52.7	26 400	81.9	41 000
3.80	54.9	27 400	85.6	42 800
3.85	57.1	28 500	89.4	44 700
3.90	59.3	29 700	93.4	46 700
3.95	61.6	30 800	97.4	48 700
4.00	64.0	32 000	102	50 800
4.05	66.4	33 200	106	52 900
4.10	68.9	34 500	110	55 200
4.15	71.5	35 700	115	57 400
4.20	74.1	37 000	120	59 800
4.25	76.8	38 400	124	62 200
4.30	79.5	39 800	129	64 600
4.35	82.3	41 200	134	67 200
4.40	85.2	42 600	140	69 800
4.45	88.1	44 100	145	72 500
4.50	91.1	45 600	150	75 200
4.55	94.2	47 100	156	78 000
4.60	97.3	48 700	162	80 900
4.65	101	50 300	168	83 900
4.70	104	51 900	174	87 000
4.75	107	53 600	180	90 100
4.80	111	55 300	187	93 300
4.85	114	57 000	193	96 600
4.90	118	58 800	200	99 900
4.95	121	60 600	207	103 000
5.00	125	62 500	214	107 000
5.10	133	66 300	228	114 000
5.20	141	70 300	244	122 000
5.30	149	74 400	260	130 000
5.40	157	78 700	276	138 000
5.50	166	83 200	294	147 000
5.60	176	87 800	312	156 000
5.70	185	92 600	331	165 000
5.80	195	97 600	351	175 000
5.90	205	103 000	371	186 000
6.00	216	108 000	392	196 000
6.50	275	137 000	513	256 000
7.00	343	172 000	656	328 000
7.50	422	211 000	826	413 000
8.00	512	256 000	1 020	512 000
8.50	614	307 000	1 250	627 000
9.00	729	365 000	1 520	758 000
9.50	857	429 000	1 820	908 000
10.0	1 000	—	2 150	—
11.0	1 330	—	2 960	—
12.0	1 730	—	3 960	—
13.0	2 200	—	5 170	—
14.0	2 740	—	6 610	—
15.0	3 380	—	8 320	—

Appendix 14 Boundary Dimensions of Radial Bearings

Units: mm

Bore Diameter	Diameter Series 8						Diameter Series 9						Diameter Series 0					
	Dimension Series						Dimension Series						Dimension Series					
	D^*	38	48	58	68	38~68	D^*	39	49	59	69	39~69	D^*	30	40	50	60	30~60
d		B				r(min)		B				r(min)		B				r(min)
0.6	2.5	1.4	—	—	—	0.05	—	—	—	—	—	—	—	—	—	—	—	—
1	3	1.5	—	—	—	0.05	4	2.3	—	—	—	0.1 (1)	—	—	—	—	—	—
1.5	4	2	—	—	—	0.05	5	2.6	—	—	—	0.15 (1)	6	3	—	—	—	0.15
2	5	2.3	—	—	—	0.08	6	3	—	—	—	0.15 (1)	7	3.5	—	—	—	0.15
2.5	6	2.6	—	—	—	0.08	7	3.5	—	—	—	0.15 (1)	8	4	—	—	—	0.15
3	7	3	—	—	—	0.1	8	4	—	—	—	0.15 (1)	9	5	—	—	—	0.15
4	9	4	—	—	—	0.1	11	5	—	—	—	0.15 (1)	12	6	—	—	—	0.2
5	11	5	—	—	—	0.15	13	6	10	—	—	0.15 (2)	14	7	—	—	—	0.2
6	13	6	—	—	—	0.15	15	7	10	—	—	0.15 (2)	17	9	—	—	—	0.3
7	14	6	—	—	—	0.15	17	7	10	—	—	0.15 (2)	19	10	—	—	—	0.3
8	16	6	8	—	—	0.2	19	9	11	—	—	0.2 (2)	22	11	14	19	25	0.3
9	17	6	8	—	—	0.2	20	9	11	—	—	0.3	24	12	15	20	27	0.3
10	19	7	9	—	—	0.3	22	10	13	16	22	0.3	26	12	16	21	29	0.3
12	21	7	9	—	—	0.3	24	10	13	16	22	0.3	28	12	16	21	29	0.3
15	24	7	9	—	—	0.3	28	10	13	18	23	0.3	32	13	17	23	30	0.3
17	26	7	9	—	—	0.3	30	10	13	18	23	0.3	35	14	18	24	32	0.3
20	32	10	12	16	22	0.3	37	13	17	23	30	0.3	42	16	22	30	40	0.6
22	34	10	—	16	22	0.3	39	13	17	23	30	0.3	44	16	22	30	40	0.6
25	37	10	12	16	22	0.3	42	13	17	23	30	0.3	47	16	22	30	40	0.6
28	40	10	—	16	22	0.3	45	13	17	23	30	0.3	52	18	24	32	43	0.6
30	42	10	12	16	22	0.3	47	13	17	23	30	0.3	55	19	25	34	45	1
32	44	10	—	16	22	0.3	52	15	20	27	36	0.6	58	20	26	35	47	1
35	47	10	12	16	22	0.3	55	15	20	27	36	0.6	62	20	27	36	48	1
40	52	10	12	16	22	0.3	62	16	22	30	40	0.6	68	21	28	38	50	1
45	58	10	13	18	23	0.3	68	16	22	30	40	0.6	75	23	30	40	54	1
50	65	12	15	20	27	0.3	72	16	22	30	40	0.6	80	23	30	40	54	1
55	72	13	17	23	30	0.3	80	19	25	34	45	1	90	26	35	46	63	1.1
60	78	14	18	24	32	0.3	85	19	25	34	45	1	95	26	35	46	63	1.1
65	85	15	20	27	36	0.6	90	19	25	34	45	1	100	26	35	46	63	1.1
70	90	15	20	27	36	0.6	100	23	30	40	54	1	110	30	40	54	71	1.1
75	95	15	20	27	36	0.6	105	23	30	40	54	1	115	30	40	54	71	1.1
80	100	15	20	27	36	0.6	110	23	30	40	54	1	125	34	45	60	80	1.1
85	110	19	25	34	45	1	120	26	35	46	63	1.1	130	34	45	60	80	1.1
90	115	19	25	34	45	1	125	26	35	46	63	1.1	140	37	50	67	90	1.5
95	120	19	25	34	45	1	130	26	35	46	63	1.1	145	37	50	67	90	1.5
100	125	19	25	34	45	1	140	30	40	54	71	1.1	150	37	50	67	90	1.5
105	130	19	25	34	45	1	145	30	40	54	71	1.1	160	41	56	75	100	2
110	140	23	30	40	54	1	150	30	40	54	71	1.1	170	45	60	80	109	2
120	150	23	30	40	54	1	165	34	45	60	80	1.1	180	46	60	80	109	2
130	165	26	35	46	63	1.1	180	37	50	67	90	1.5	200	52	69	95	125	2
140	175	26	35	46	63	1.1	190	37	50	67	90	1.5	210	53	69	95	125	2
150	190	30	40	54	71	1.1	210	45	60	80	109	2	225	56	75	100	136	2.1
160	200	30	40	54	71	1.1	220	45	60	80	109	2	240	60	80	109	145	2.1
170	215	34	45	60	80	1.1	230	45	60	80	109	2	260	67	90	122	160	2.1
180	225	34	45	60	80	1.1	250	52	69	95	125	2	280	74	100	136	180	2.1
190	240	37	50	67	90	1.5	260	52	69	95	125	2	290	75	100	136	180	2.1
200	250	37	50	67	90	1.5	280	60	80	109	145	2.1	310	82	109	150	200	2.1
220	270	37	50	67	90	1.5	300	60	80	109	145	2.1	340	90	118	160	218	3
240	300	45	60	80	109	2	320	60	80	109	145	2.1	360	92	118	160	218	3
260	320	45	60	80	109	2	360	75	100	136	180	2.1	400	104	140	190	250	4
280	350	52	69	95	125	2	380	75	100	136	180	2.1	420	106	140	190	250	4
300	380	60	80	109	145	2.1	420	90	118	160	218	3	460	118	160	218	290	4
320	400	60	80	109	145	2.1	440	90	118	160	218	3	480	121	160	218	290	4
340	420	60	80	109	145	2.1	460	90	118	160	218	3	520	133	180	243	325	5
360	440	60	80	109	145	2.1	480	90	118	160	218	3	540	134	180	243	325	5
380	480	75	100	136	180	2.1	520	106	140	190	250	4	560	135	180	243	325	5
400	500	75	100	136	180	2.1	540	106	140	190	250	4	600	148	200	272	355	5
420	520	75	100	136	180	2.1	560	106	140	190	250	4	620	150	200	272	355	5
440	540	75	100	136	180	2.1	600	118	160	218	290	4	650	157	212	280	375	6

Notes (1) Chamfer dimensions for dimension series **39** only.

(2) For dimension series **39**, chamfer dimensions are **0.2** mm for bore diameters of **5** and **6** mm and **0.3** mm for bore diameters of **7** and **8** mm.

∗ : Outside Diameter

Appendix 15 Boundary Dimensions of Thrust Bearings

Units: mm

Bore Diameter	Diameter Series 0					Diameter Series 1					Diameter Series 2				
	Dimension Series					Dimension Series					Dimension Series				
	D^*	70	90	10	r(min)	D^*	71	91	11	r(min)	D^*	72	92	12	r(min)
d		H					H					H			
4	12	4	—	6	0.3	—	—	—	—	—	16	6	—	8	0.3
6	16	5	—	7	0.3	—	—	—	—	—	20	6	—	9	0.3
8	18	5	—	7	0.3	—	—	—	—	—	22	6	—	9	0.3
10	20	5	—	7	0.3	24	6	—	9	0.3	26	7	—	11	0.6
12	22	5	—	7	0.3	26	6	—	9	0.3	28	7	—	11	0.6
15	26	5	—	7	0.3	28	6	—	9	0.3	32	8	—	12	0.6
17	28	5	—	7	0.3	30	6	—	9	0.3	35	8	—	12	0.6
20	32	6	—	8	0.3	35	7	—	10	0.3	40	9	—	14	0.6
25	37	6	—	8	0.3	42	8	—	11	0.6	47	10	—	15	0.6
30	42	6	—	8	0.3	47	8	—	11	0.6	52	10	—	16	0.6
35	47	6	—	8	0.3	52	8	—	12	0.6	62	12	—	18	1
40	52	6	—	9	0.3	60	9	—	13	0.6	68	13	—	19	1
45	60	7	—	10	0.3	65	9	—	14	0.6	73	13	—	20	1
50	65	7	—	10	0.3	70	9	—	14	0.6	78	13	—	22	1
55	70	7	—	10	0.3	78	10	—	16	0.6	90	16	21	25	1
60	75	7	—	10	0.3	85	11	—	17	1	95	16	21	26	1
65	80	7	—	10	0.3	90	11	—	18	1	100	16	21	27	1
70	85	7	—	10	0.3	95	11	—	18	1	105	16	21	27	1
75	90	7	—	10	0.3	100	11	—	19	1	110	16	21	27	1
80	95	7	—	10	0.3	105	11	—	19	1	115	16	21	28	1
85	100	7	—	10	0.3	110	11	—	19	1	125	18	24	31	1
90	105	7	—	10	0.3	120	14	—	22	1	135	20	27	35	1.1
100	120	9	—	14	0.6	135	16	21	25	1	150	23	30	38	1.1

∗ : Outside Diameter

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